

Youngdong Jang

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EDUCATION

- Korea University, Seoul, Republic of Korea Mar. 2023 - Feb. 2028 (Expected)
Ph.D. Candidate - Artificial Intelligence
- Dongguk University, Seoul, Republic of Korea Mar. 2015 - Feb. 2021
B.S. - Mathematics

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, *=EQUAL CONTRIBUTION

- [C5] Gyeongrok Oh, **Youngdong Jang**, Jonghyun Choi, Suk-Ju Kang, Guang Lin, Sangpil Kim.,
Training-Free Global Geometric Association for 4D LiDAR Panoptic Segmentation,
Pre-print
- [C4] **Youngdong Jang***, Gyeongrok Oh*, Jong Wook Kim, Hyunju Ryu, Hyung-gun Chi, SeungHyeon Kim,
Seungryong Kim, Jonghyun Choi, Sangpil Kim.,
Motion Cues from Image-based Point Tracking for LiDAR Scene Flow Estimation,
The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS 2026)
- [C3] Sumin In, **Youngdong Jang**, Utae Jeong, Minhyuk Jang, Heongcheol Park, Eunbyung Park, Sangpil
Kim.,
CompMarkGS: Robust Watermarking for Compressed 3D Gaussian Splatting,
The Fourteenth International Conference on Learning Representations (ICLR 2026)
- [C2] **Youngdong Jang**, Hyunje Park, Feng Yang, Heeju Ko, Euijin Choo, Sangpil Kim.,
3D-GSW: 3D Gaussian Splatting Watermark for Robust Watermarking,
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2025
- [C1] **Youngdong Jang**, Dongin Lee, Minhyuk Jang, Jongwook Kim, Feng Yang, Sangpil Kim.,
WaterF: Robust Watermarks in Radiance Fields for Protection of Copyrights,
The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2024
- [J2] **Youngdong Jang***, Minhyuk Jang*, JaeHyeok Lee, Feng Yang, Gyeongrok Oh, Jongheon Jeong, Sangpil
Kim.,
LVMark: Robust Watermark for Latent Video Diffusion Models,
IEEE Transactions on Information Forensics and Security (TIFS), JCR Top 10%
- [J1] Minhyuk Jang, Jongwook Kim, **Youngdong Jang**, Donghyuk Kim, Wonseok Roh, Inyong Hwang,
Guang Lin, Sangpil Kim.,
High-Quality Three-Dimensional Cartoon Avatar Reconstruction with Gaussian Splatting,
Engineering Applications of Artificial Intelligence (EAAI), JCR Top 10%

EXPERIENCE

- **Structures-Computer Interaction Lab @ University of California, Los Angeles** Jun. 2026 - Present
Advisor: M. Khalid Jawed
 - Develop an autonomous data acquisition pipeline using mobile robots equipped with RGB-D cameras and LiDAR sensors
 - Research 3D reconstruction algorithms to convert sparse robotic sensor data into high-fidelity 3D models
- **Computer Vision Lab @ Korea University** Mar. 2023 - Present
Advisor: Sangpil Kim
 - [C5] 4D Panoptic Segmentation (with **Hyundai Motor** & **Sogang Univ.**)
 - [C4] LiDAR Scene Flow Estimation (with **Hyundai Motor** & **Purdue Univ.** & **KAIST**)
 - [C3] Integrate quantization and watermarking into 3D Gaussian Splatting (with **Yonsei Univ.**)
 - [C2] Watermarking of 3DGS (with **Google DeepMind** & **Univ. of Alberta**)
 - [C1] Watermarking of NeRF-based model (with **Google Research**)

- [J2] Watermarking of Video Diffusion model (with **Google DeepMind**)
- [J1] 3DGS-based human reconstruction (with **Hanwha Systems & Purdue Univ.**)
- **Scalar Data, Inc.** Jun. 2022 - Nov. 2022
Data Engineer
 - Enterprise data management for EV Modu, an all-in-one electric vehicle charging platform
 - Build Electricity Demand Response (DR) system with Korea Power Exchange

ACADEMIC SERVICE

- **Reviewer**
 - IEEE Transactions on Information Forensics and Security, IEEE Transactions on Multimedia, IEEE Transactions on Image Processing, ACM Transactions on Graphics, NeurIPS 2025 2026, ECCV 2026, CVPR 2025 2026, SIGGRAPH Asia 2026, AAAI 2027
- **Teaching Assistant**
 - [2025-2] Copyright Protection In AI (XAI636)

PATENTS & TECHNOLOGY TRANSFERS

- **Patent Applications**
 - Sangpil Kim, Youngdong Jang, "Method And Apparatus For Fine-Tuning 3D Artificial Intelligence Model To Embed Watermarking Capabilities Using Discrete Wavelet Transform," Korean Patent Application No.10-2024-0167666 (Filed: Nov.2024).
- **Technology Transfers**
 - Transferred core technical know-how on 3D AI models to NAWA
 - Transferred core technical know-how on 3D content copyright protection to AIWORKX

SCHOLARSHIPS & AWARDS

- AI Seoul Tech Scholarship @ Seoul Scholarship Foundation Nov. 2025
- Merit Scholarship @ Department of Mathematics, Dongguk University Sep. 2020
- Merit Scholarship @ Department of Mathematics, Dongguk University Mar. 2020